Prerequisites

Understanding of the following services….. Jus know what each service mentioned here, does
aws cloud services-foundational— to know abt aws fundamental services
Automate Cloud Audits – AWS Audit Manager
What Is AWS CloudTrail? - AWS CloudTrail
What is Amazon CloudWatch? - Amazon CloudWatch
What is IAM? - AWS Identity and Access Management
AWS Security Hub
AWS Cloudtrail Insights vs AWS Macie vs AWS GuardDuty - Stack Overflow

Thoroughly do the following sagemaker courses to understand ml pipeline on aws
https://explore.skillbuilder.aws/learn/courses/240/introduction-to-amazon-sagemaker
https://explore.skillbuilder.aws/learn/courses/22175/amazon-sagemaker-ai-getting-started
https://explore.skillbuilder.aws/learn/courses/17719/online-course-supplement-practical-data-science-with-amazon-sagemaker

AI Tools and Services – Artificial Intelligence Products – AWS

Understanding of the following services….. Jus know what each service mentioned here, does
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https://explore.skillbuilder.aws/learn/courses/17719/online-course-supplement-practical-data-science-with-amazon-sagemaker

AI Tools and Services – Artificial Intelligence Products – AWS

https://www.examtopics.com/exams/amazon/aws-certified-generative-ai-developer-professional-aip-c01/

Quick scenario based understanding

AWS Generative AI Developer Exam - Services Explained Through One Real Company Scenario

Imagine you work for **ABC Insurance Company** building an AI-powered insurance assistant using Amazon Bedrock.

A customer uploads insurance documents and asks:

> "Can you summarize my policy and tell me whether my accident is covered?"

Let's see how each AWS service participates.

1. Amazon Cognito (User Login)

Real Scenario

Customer Priya opens the insurance portal.

Before accessing the AI chatbot, she must sign in.

What Cognito Does

- User registration
- Login
- Password reset
- Google Login
- Facebook Login

Flow

Customer
↓
Cognito Login
↓
Authenticated User

Exam Tip

If question says:

> Users need to authenticate before accessing a GenAI application.

Answer:
Cognito

Pricing

- Large free tier
- Then pay per monthly active user

Easy Memory

Cognito = Login System

2. API Gateway

Real Scenario

Customer asks:

> "Summarize my policy"

Frontend must send request somewhere.

API Gateway receives the request.

Flow

Browser
↓
API Gateway
↓
Lambda

Why Use It?

- Creates APIs
- Handles millions of requests
- Security
- Rate limiting

Exam Tip

Need REST API?

API Gateway

Pricing

Pay per API request.

Easy Memory

API Gateway = Receptionist of your application

3. AWS Lambda

Real Scenario

Request reaches Lambda.

Lambda:

1. Reads uploaded PDF
2. Extracts text
3. Sends text to Bedrock

Flow

PDF Uploaded
↓
Lambda
↓
Bedrock

Why Important?

No server management.

AWS automatically runs code.

Example

Upload claim document.

Lambda immediately starts processing.

Pricing

- First 1 million requests free
- Then pay per execution time

Easy Memory

Lambda = Worker who performs tasks

4. Amazon S3

Real Scenario

Customer uploads:

- Insurance policy
- Claim form
- Accident report

These are stored in S3.

Flow

Customer
↓
S3 Bucket

GenAI Usage

Knowledge Base documents usually come from S3.

Pricing

Pay for:

- Storage
- Requests

Easy Memory

S3 = Filing Cabinet

5. Amazon Bedrock ⭐⭐⭐

Real Scenario

Lambda sends document text.

Prompt:

> Summarize this policy and explain coverage.

Bedrock invokes Claude.

Claude returns answer.

Flow

Lambda
↓
Bedrock
↓
Claude

Services Inside Bedrock

- Foundation Models
- Guardrails
- Agents
- Prompt Management
- Knowledge Bases
- Prompt Caching
- Model Evaluation

Exam Questions Often Ask

Which service provides access to foundation models?

Bedrock

Pricing

Pay for:

- Input Tokens
- Output Tokens

Easy Memory

Bedrock = AI Brain

6. Amazon Q Business

Real Scenario

Employees ask:

> What is maternity leave policy?
>
> What is the reimbursement process?

Q Business searches:

- PDFs
- SharePoint
- Confluence

and answers.

Flow

Company Documents
↓
Q Business
↓
Employee Chat

Exam Tip

Internal enterprise chatbot?

Amazon Q Business

Easy Memory

Q Business = ChatGPT for company documents

7. Amazon Q Developer

Real Scenario

Developer asks:

> Generate Lambda code to invoke Claude.

Q Developer writes code.

Helps With

- Coding
- Documentation
- Debugging
- Unit Tests

Exam Tip

Developer productivity assistant?

Amazon Q Developer

Easy Memory

Q Developer = AI Pair Programmer

8. Step Functions

Real Scenario

Claim document processing.

Steps:

1. Upload PDF
2. Extract Text
3. Bedrock Summary
4. Fraud Detection
5. Save Result
6. Notify Customer

Flow

Upload
↓
Extract
↓
Summarize
↓
Fraud Check
↓
Store
↓
Notify

Why Not Lambda Alone?

Many steps.

Need orchestration.

Exam Tip

Multi-step workflow?

Step Functions

Pricing

Pay per state transition.

Easy Memory

Step Functions = Project Manager

9. Amazon SQS

Real Scenario

10,000 customers upload claims.

Lambda cannot process all immediately.

SQS stores requests.

Flow

Users
↓
SQS Queue
↓
Lambda

Benefits

- Prevent overload
- Reliable processing

Exam Tip

Need decoupling?

SQS

Easy Memory

SQS = Waiting Line

10. Amazon SNS

Real Scenario

Claim processing completed.

Customer gets email:

> Your claim summary is ready.

Flow

Bedrock Complete
↓
SNS
↓
Email

Can Send

- Email
- SMS
- Mobile notifications

Exam Tip

Notification service?

SNS

Easy Memory

SNS = Announcement System

11. CloudWatch

Real Scenario

Chatbot suddenly becomes slow.

CloudWatch shows:

- Latency
- Errors
- CPU
- Logs

Flow

Lambda
Bedrock
API Gateway
↓
CloudWatch

Exam Tip

Monitoring?

CloudWatch

Easy Memory

CloudWatch = CCTV Camera

12. CloudTrail

Real Scenario

Security team asks:

> Who deleted the Bedrock Guardrail?

CloudTrail answers.

Tracks

- API Calls
- Console Actions
- Resource Changes

Exam Tip

Audit trail?

CloudTrail

Easy Memory

CloudTrail = Security Guard Logbook

13. IAM

Real Scenario

Lambda needs permission to invoke Bedrock.

Without IAM:

❌ Access Denied

Policy Example

```
{
"Action":"bedrock:InvokeModel"
}
```

Exam Tip

Permissions?

IAM

Easy Memory

IAM = Security Guard

14. IAM Identity Center

Real Scenario

Company has:

- Dev Account
- Test Account
- Production Account

Employee logs in once.

Accesses all accounts.

Exam Tip

SSO?

IAM Identity Center

Easy Memory

Identity Center = Master Key

15. AWS Config

Real Scenario

Company policy:

> No public S3 buckets allowed.

Someone makes bucket public.

AWS Config detects violation.

Exam Tip

Compliance monitoring?

AWS Config

Easy Memory

AWS Config = Compliance Auditor

16. Amazon Macie

Real Scenario

Customer uploads claim documents.

Macie discovers:

- PAN card
- Aadhaar
- Credit card numbers

Flow

S3
↓
Macie Scan
↓
PII Found

Why Important?

Prevent sensitive data leaks to AI systems.

Exam Tip

Sensitive data discovery?

Macie

Easy Memory

Macie = Data Detective

17. Glue DataBrew

Real Scenario

Training data contains:

Name	Age
Priya	Null
Ketan	35

Before training model:

DataBrew cleans data.

Features

- Remove duplicates
- Fill missing values

- Normalize columns

Exam Tip

No-code data preparation?

DataBrew

Easy Memory

DataBrew = Data Cleaning Maid

18. SageMaker

Real Scenario

Insurance company wants its own fraud prediction model.

Historical claims:

- Fraud
- Genuine

Train custom ML model.

Flow

Training Data
↓
SageMaker
↓
Fraud Model

Bedrock vs SageMaker

Bedrock	SageMaker
Use FM	Build ML
Claude	XGBoost
GenAI	Traditional ML

Pricing

Pay for:

- Notebook
- Training Jobs
- Endpoints

Easy Memory

SageMaker = ML Factory

Complete Exam Architecture

User
↓
Cognito
↓
API Gateway
↓
Lambda
↓
Bedrock
↓
Claude

Documents
↓
S3
↓
Knowledge Base

High Traffic
↓
SQS

Notifications
↓
SNS

Monitoring

↓
CloudWatch

Auditing
↓
CloudTrail

Permissions
↓
IAM

Compliance
↓
Config

PII Detection
↓
Macie

Workflow
↓
Step Functions

Super-Important Exam Mapping

Requirement	Service
Foundation Models	Bedrock
Enterprise Chatbot	Q Business
Coding Assistant	Q Developer
User Login	Cognito
Permissions	IAM
SSO	Identity Center
Workflow	Step Functions
Queue	SQS
Notifications	SNS
Monitoring	CloudWatch

Auditing	CloudTrail
Compliance	Config
PII Detection	Macie
Data Cleaning	DataBrew
Custom ML	SageMaker
Serverless Compute	Lambda
APIs	API Gateway

So finally,

AWS Services for Generative AI Developer Exam

AWS Lambda

What is Lambda?

AWS Lambda is a serverless compute service that allows you to run code without managing servers.

You simply upload your code, and AWS automatically runs it whenever an event occurs.

Why Use Lambda?

- No server management
- Automatic scaling
- Pay only when code runs
- Integrates with many AWS services

Example

A user uploads a PDF to Amazon S3.

Lambda automatically:

- Reads the PDF
- Extracts text
- Sends the content to Amazon Bedrock for summarization

Common GenAI Use Cases

- Invoke Bedrock models
- Process uploaded documents
- Trigger AI workflows
- Build serverless chatbots

Pricing

- First 1 million requests per month are free
- Pay based on number of requests and execution duration

Exam Keywords

Serverless Compute, Event-Driven Architecture

Amazon API Gateway

What is API Gateway?

API Gateway is a managed service for creating and managing APIs.

It acts as the entry point for applications.

Why Use API Gateway?

- Create REST APIs
- Secure APIs
- Rate limiting
- Monitoring

Example

A chatbot user sends a message.

The request goes through API Gateway, which forwards it to Lambda and Bedrock.

Common GenAI Use Cases

- Chatbot APIs
- AI-powered applications
- Mobile app backends

Pricing

Pay per API request.

Exam Keywords

REST API, HTTP API, WebSocket API

AWS Step Functions

What is Step Functions?

Step Functions is a workflow orchestration service that coordinates multiple AWS services.

Why Use Step Functions?

Instead of writing complex code to manage workflows, Step Functions manages the process for you.

Example

Document Processing Workflow:

1. Upload PDF
2. Extract Text
3. Summarize with Bedrock
4. Store Results
5. Notify User

Common GenAI Use Cases

- RAG pipelines
- Document processing
- Agent workflows
- Approval systems

Pricing

Pay per state transition.

Exam Keywords

Workflow Automation, State Machine, Orchestration

Amazon Macie

What is Macie?

Amazon Macie discovers and protects sensitive data stored in Amazon S3.

What Can It Detect?

- Credit card numbers
- Passport numbers
- PAN cards
- Aadhaar numbers
- Personally Identifiable Information (PII)

Example

Before documents are sent to Bedrock, Macie scans them for sensitive information.

Common GenAI Use Cases

- Data privacy
- Compliance
- PII detection before AI processing

Pricing

Based on amount of S3 data scanned.

Exam Keywords

PII Detection, Data Privacy, Sensitive Data Discovery

AWS Glue DataBrew

What is DataBrew?

DataBrew is a visual data preparation service.

It allows users to clean and transform data without coding.

Features

- Remove duplicates
- Fill missing values
- Standardize formats
- Data profiling

Example

Training data contains missing values and inconsistencies.

DataBrew cleans the data before model training.

Common GenAI Use Cases

- Preparing training datasets
- Cleaning data for analytics
- Data preprocessing

Pricing

Pay per job execution.

Exam Keywords

Data Cleaning, No-Code Data Preparation

IAM Identity Center

What is IAM Identity Center?

IAM Identity Center provides Single Sign-On (SSO) across AWS accounts and applications.

Benefits

- One login for multiple AWS accounts

- Centralized access management
- Improved security

Example

A company has Development, Testing, and Production accounts.

Employees log in once and access all accounts.

Pricing

No additional charge.

Exam Keywords

Single Sign-On, SSO

AWS Identity and Access Management (IAM)

What is IAM?

IAM controls access to AWS resources.

Components

Users
Individual identities.

Groups
Collection of users.

Roles
Temporary permissions.

Policies
Permission rules.

Example

A Lambda function needs permission to invoke Bedrock.

IAM grants that permission.

Common GenAI Use Cases

- Allow Bedrock access
- Control SageMaker permissions
- Secure AI applications

Pricing

Free.

Exam Keywords

Authentication, Authorization, Least Privilege

AWS Config

What is AWS Config?

AWS Config records and evaluates AWS resource configurations.

What Does It Do?

- Tracks configuration changes
- Maintains resource history
- Evaluates compliance rules

Example

A company policy states that no S3 bucket should be public.

AWS Config continuously checks compliance.

Pricing

Pay per configuration item recorded.

Exam Keywords

Compliance, Governance, Configuration Tracking

Amazon Cognito

What is Cognito?

Amazon Cognito provides authentication and user management for applications.

Features

- User sign-up
- User sign-in
- Password recovery
- Social login

Supported Identity Providers

- Google
- Facebook
- Apple
- Enterprise providers

Example

Users must log in before accessing an AI chatbot.

Cognito manages the authentication process.

Pricing

Generous free tier followed by monthly active user pricing.

Exam Keywords

Authentication, User Management

Amazon Bedrock

What is Bedrock?

Amazon Bedrock is a fully managed service that provides access to foundation models through APIs.

Available Models

- Claude
- Amazon Nova
- Llama
- Mistral
- Stable Diffusion

Major Features

Foundation Models
Access leading AI models.

Knowledge Bases
Implement Retrieval Augmented Generation (RAG).

Agents
Automate multi-step tasks.

Guardrails
Apply safety controls.

Prompt Management
Store and version prompts.

Model Evaluation
Compare foundation models.

Prompt Caching
Reduce latency and cost.

Example

A user uploads a PDF and asks for a summary.

Bedrock uses a foundation model to generate the response.

Pricing

Pay for input and output tokens.

Pricing varies by model.

Exam Keywords

Foundation Models, Generative AI, Prompt Engineering, RAG

Amazon SageMaker

What is SageMaker?

Amazon SageMaker is AWS's machine learning platform for building, training, and deploying models.

Features

- Data preparation
- Model training
- Hyperparameter tuning
- Deployment
- Monitoring

Example

A company wants to build a fraud detection model using historical claims data.

SageMaker is used to train and deploy the model.

Bedrock vs SageMaker

Bedrock is used when you want to use existing foundation models.

SageMaker is used when you want to build and train your own machine learning models.

Pricing

Pay for notebooks, training jobs, and endpoints.

Exam Keywords

Machine Learning Platform, Model Training

Amazon SQS

What is SQS?

Amazon Simple Queue Service (SQS) is a managed message queue service.

Why Use It?

- Decouple applications
- Handle traffic spikes
- Improve reliability

Queue Types

Standard Queue
High throughput.

FIFO Queue
First-In-First-Out processing.

Example

Thousands of document processing requests arrive simultaneously.

SQS stores requests until workers process them.

Pricing

Pay per request.

Exam Keywords

Queue, Decoupling, Messaging

Amazon SNS

What is SNS?

Amazon Simple Notification Service (SNS) is a publish-subscribe messaging service.

Supported Targets

- Email
- SMS
- Lambda
- HTTP endpoints

Example

When a Bedrock summarization job completes, SNS sends a notification email.

Common Uses

- Alerts
- Notifications
- Event broadcasting

Pricing

Pay per message delivered.

Exam Keywords

Publish-Subscribe, Notification Service

Amazon CloudWatch

What is CloudWatch?

CloudWatch is AWS's monitoring and observability service.

What Does It Monitor?

- Metrics
- Logs
- Alarms
- Dashboards

Example

Monitor chatbot response time, Lambda errors, and Bedrock latency.

Pricing

Free tier available.
Additional charges for logs, metrics, and alarms.

Exam Keywords

Monitoring, Metrics, Logging

AWS CloudTrail

What is CloudTrail?

CloudTrail records AWS account activity and API calls.

What Does It Track?

- Console actions
- API requests
- Resource changes

Example

An administrator deletes a Bedrock Guardrail.

CloudTrail records who performed the action and when.

Pricing

Management event logging is included.
Additional features may incur charges.

Exam Keywords

Auditing, Governance, API Tracking

Amazon Q Developer

What is Amazon Q Developer?

Amazon Q Developer is an AI assistant for software developers.

Capabilities

- Generate code
- Explain code

- Debug code
- Create documentation
- Generate tests

Example

A developer asks:

"Create Python code to invoke a Bedrock model."

Amazon Q Developer generates the code.

Pricing

Free tier available.
Professional plans available.

Exam Keywords

AI Coding Assistant, Developer Productivity

Amazon Q Business

What is Amazon Q Business?

Amazon Q Business is an enterprise AI assistant that answers questions using company data.

Supported Data Sources

- SharePoint
- Confluence
- Salesforce
- S3
- Google Drive

Capabilities

- Search enterprise content
- Summarize documents
- Answer employee questions

Example

An employee asks:

"What is our leave policy?"

Q Business searches company documents and generates an answer.

Pricing

Typically based on users and usage.

Exam Keywords

Enterprise AI Assistant, Knowledge Discovery

Final Exam Revision Table

Service	Main Purpose
Lambda	Run code without servers
API Gateway	Create and manage APIs
Step Functions	Orchestrate workflows
Macie	Discover sensitive data
DataBrew	Clean and prepare data
IAM Identity Center	Single Sign-On
IAM	Manage permissions
AWS Config	Compliance monitoring
Cognito	User authentication
Bedrock	Foundation model platform
SageMaker	Machine learning platform
SQS	Message queue
SNS	Notifications
CloudWatch	Monitoring and logging

CloudTrail	Auditing and tracking
Q Developer	AI coding assistant
Q Business	Enterprise AI assistant

A good way to remember them is:

Cognito authenticates users, API Gateway receives requests, Lambda executes logic, Bedrock generates AI responses, IAM secures access, SQS handles load, SNS sends notifications, CloudWatch monitors, CloudTrail audits, Config checks compliance, Macie protects sensitive data, and SageMaker builds custom ML models.

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